Quick guide CESGA

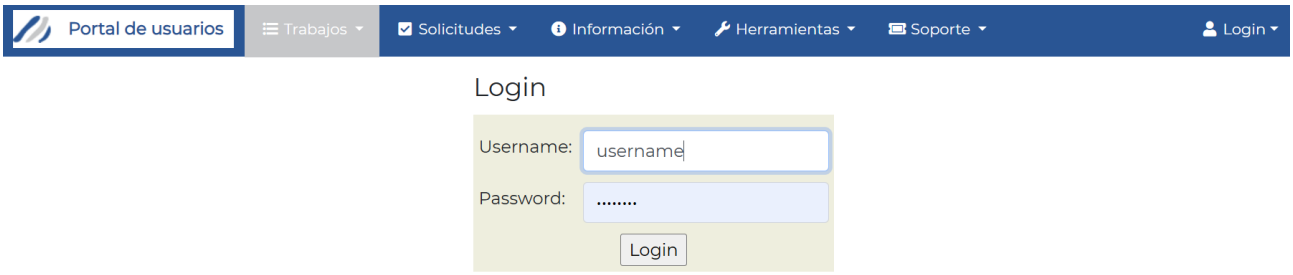
Deep Learning: Eduardo Mosqueira Rey and Samuel Magaz Romero

Academic year 2025/2026

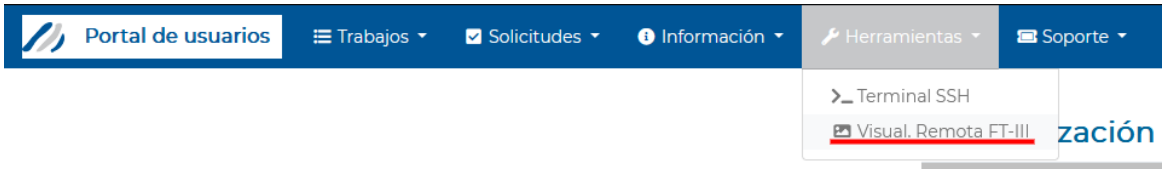
Brief guide on how to make use of CESGA resources for Deep Learning practices.
***NOTE**: it is assumed that the user is registered with CESGA.*

1 Instructions

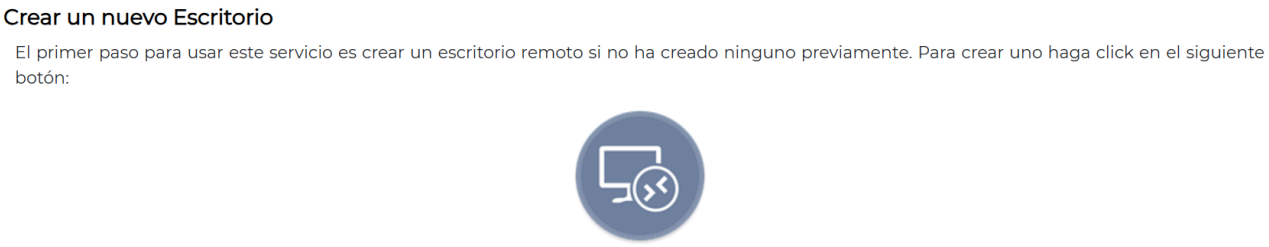
1. You can find the CESGA Technical Documentation here https://cesga-docs.gitlab.io/index.html. One good point to start is the First Steps.
2. Go to https://portalusuarios.cesga.es and enter your login and password.



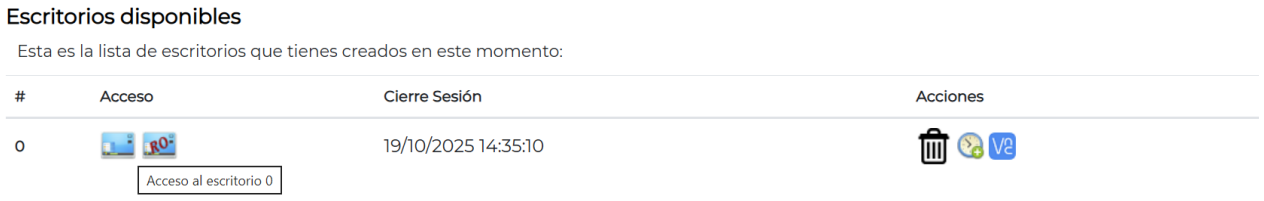
3. Start a remote viewing session in Finisterrae-III (through the "Visual Remota FT-III' menu option).



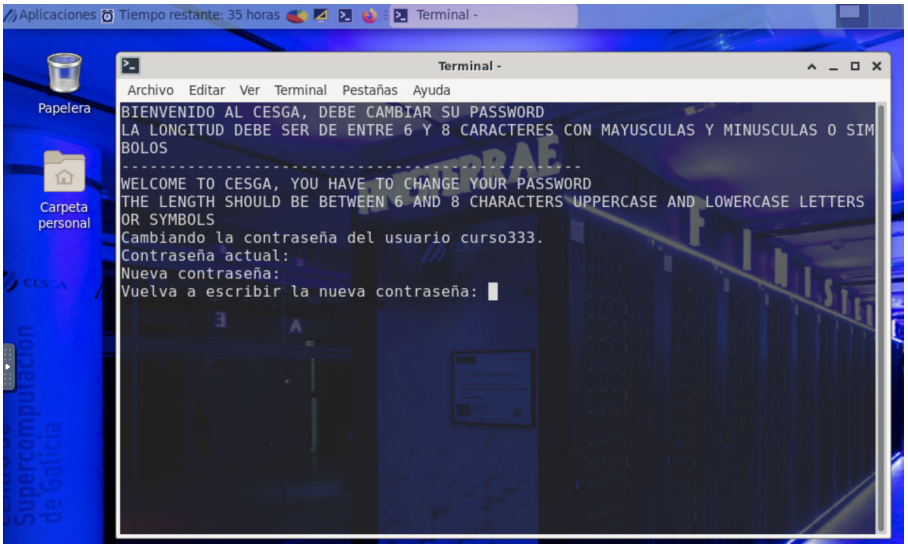
4. At first create a remote desktop



5. Access the remote desktop



6. Enter the remote desktop and open a terminal, the first time you'll be asked to change your password.



7. To transfer files to your account you can use a SFTP client (e.g. FileZilla) using the following information (see more details in the documentation: `https://cesga-docs.gitlab.io/ft3-user-guide/data_transfer.html#data-transfer`).

```
Protocol: SFTP
Name or server IP: ft3.cesga.es
Port: 22
Username: username (just the name, not the full email)
Password: your_password
```

8. Take into account that, if you are accessing CESGA's resources from an authorized center, you do not need to configure the VPN. In case you are accessing from a non-authorized center (e.g. your home) you will need to set up a VPN. You have all the details about setting up a VPN client and accessing CESGA servers in the following documentation: `https://cesga-docs.gitlab.io/ft3-user-guide/how_to_connect.html`.

9. In the terminal, request a compute session with GPU and 16GB of memory (depending on the project it will be necessary to request more or less memory):

```
compute --gpu --mem 16G
```

   This computing session will have access to the different files of our user. Due to storage quotas, it may be necessary to follow these steps in order to install all the necessary packages for our environment (it is recommended to do it first, in order to avoid problems later on).

10. Load the module `miniconda3` provided by CESGA.

```
module load cesga/system miniconda3
```

11. Create and activate a new Conda environment:

```
conda create --name mia-dl python=3.11
conda activate mia-dl
```

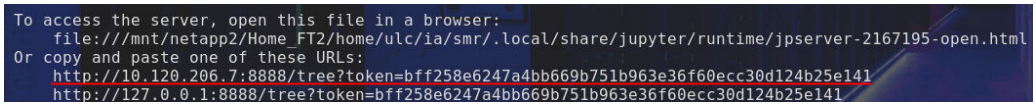
12. Install the necessary packages for the practice:

```
pip install tensorflow[and-cuda]
pip install keras
pip install matplotlib
pip install notebook
```

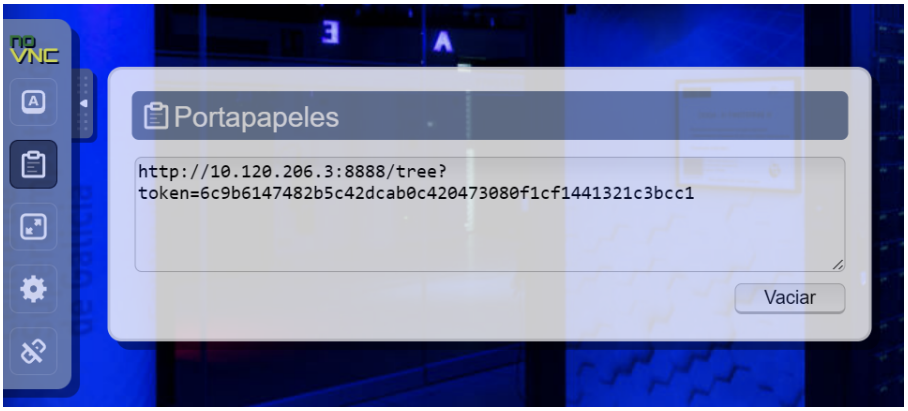
13. Launch the `jupyter` environment so that we can access it from our remote virtualization session (the environment runs on the machine corresponding to the computing session):

```
jupyter notebook --no-browser --ip=$(hostname -i)
```

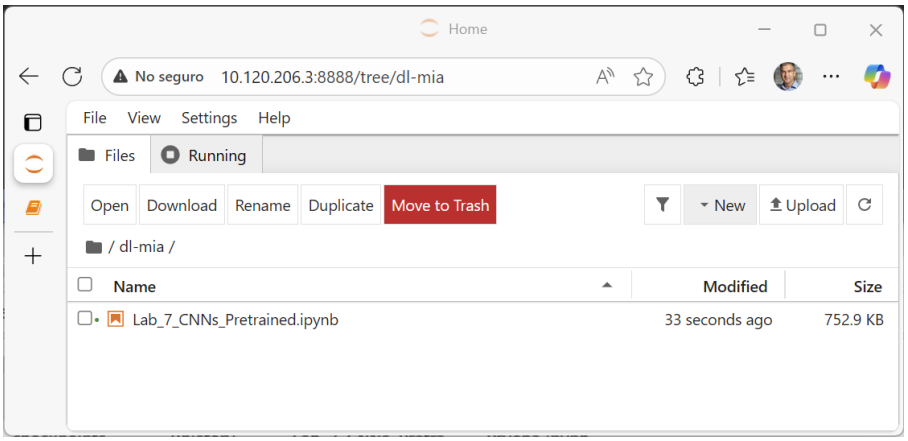
In the console, the environment execution trace will start to be displayed, and when it is loaded, the address (underlined in the Figure) where we can access the web application will appear:



14. For copying and pasting between your local PC and the remote desktop you can use the left panel application, that is used to facilitate communication between your local PC and the remote desktop. It also includes a keyboard, a full-screen mode, settings and a disconnection icon.

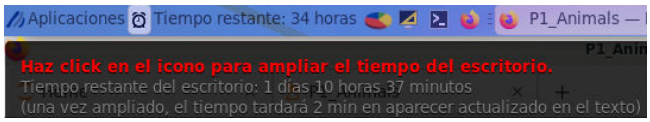


15. Pasting the address given into a local browser allows you to open your remote notebooks in the local machine.



2 Considerations

- **Lifetime**:
  - Computing sessions have a maximum lifetime of 8 hours, after which they are closed directly, regardless of whether any task is being performed or not. It is important to organize the work with this in mind, as it could happen that the session is closed in the middle of a training session.
  - Remote viewing sessions have a default lifetime of 36 hours, which can be manually increased from within the environment.



- **Execution on CPU or GPU**:
  - The resources available to CESGA are abundant but also limited, so it may take some time to be assigned with a GPU when requesting the computation session.
  - This guide is a reference to be able to use these resources, but it is up to the user to choose whether to perform the tasks with CPU (always available with the remote virtualization session) or with GPU (available in a limited way with a maximum of 8 hours per session).